

Beyond Seen Mistakes: Diagnosing Natural Error-Composition Shift in Multi-Criterion Exercise Assessment

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Abstract—Automated exercise assessment predicts several biomechanical criteria at once, but ordinary grouped evaluation does not directly test a diagnosis composition chosen to be absent from training. We construct a recording-pair-disjoint observed-composition holdout protocol on the public ALEX-GYM-1 pose data. The protocol audits missing detections, excludes one unrecoverable two-view repetition, and retains 12 natural diagnosis compositions whose training and validation folds contain both states of every criterion. Across five sequence models, exact diagnosis match is 11.34 percentage points lower on targeted leave-one-composition-out tests than on ordinary grouped validation (target bootstrap 95% CI [8.09, 14.54]; two-sided Holm-adjusted $p = .00098$). We then define Label-Opposition Pressure (LOP), the number of training examples that match a held-out diagnosis on every other criterion while carrying the opposite current bit. Within diagnosis, larger LOP is strongly associated with lower criterion accuracy (raw diagnosis-equal mean Spearman $\rho = -.651$; 50,000-shuffle blocked $p < .00004$). The result is a reproducible stress test and a training-label-only diagnostic of natural composition failure, not a claim of universal architecture superiority or causal label influence.

Index Terms—exercise assessment, multi-label classification, compositional generalization, skeleton sequences, label-space diagnostics

I. INTRODUCTION

Fine-grained exercise assessment differs from ordinary action recognition. A system must report several simultaneous errors—for example foot orientation, hip position, back posture, and movement depth—rather than one action class. Consequently, a label is a binary diagnosis vector $\mathbf{y} \in \{0,1\}^C$, and the practically relevant question is whether individually familiar criterion states remain recognizable when they occur in a new combination.

The distinction is easy to miss. A recording-disjoint random split protects against direct source overlap, but it does not deliberately hold out a chosen diagnosis vector. A model can therefore obtain reasonable marginal accuracy while using pose patterns correlated with entire diagnoses. This matters for feedback: one systematically wrong criterion invalidates an otherwise plausible diagnosis.

Composite errors have been studied directly by CPR-Coach, which trains on isolated error classes and tests synthetically enumerated composites [2]. Held-out label combinations are also established in multi-label learning [3]. Our setting is narrower and naturalistic: training remains multi-error, every

evaluated criterion state has training support, and test diagnoses are naturally observed in a public two-view exercise dataset. We do not claim the first composite-error task or the first combination split.

We organize the study around two retrospective research questions. **RQ1** asks whether ordinary grouped validation hides failure on a diagnosis deliberately absent from training. **RQ2** asks whether the local training-label neighborhood identifies the vulnerable criteria. Because these questions were fixed after an initial repository audit rather than preregistered, all primary tests are two-sided and Holm-corrected.

We contribute a data-audited, recording-pair-disjoint observed-composition protocol with explicit feasibility rules and a frozen split manifest. We also introduce a training-label-only opposition statistic that strongly localizes criterion failure within held-out diagnoses. Figure 1 shows the sparse, naturally observed diagnosis spaces and the targets retained by the final support rule.

II. RELATED WORK

A. Exercise assessment and composite errors

ALEX-GYM-1 provides two-view 3D poses and multi-criterion exercise annotations [1]. Other rehabilitation datasets emphasize clinical scores, correctness, or multi-task assessment rather than natural held-out multi-error label sets [4]–[6]. FineDiving and FineParser study procedure-aware action quality at fine temporal granularity [7], [8], but not this diagnosis-set shift. CPR-Coach is the closest direct composition formulation [2]; its isolated-error training regime differs from our naturally multi-error training pool.

Learnable Motion-Focused Tokenization (LMFT) drops low-motion RGB patches to reduce background-driven video domain shift [11]. Our normalized pose input contains no scene background, while static joint configurations can define form errors. LMFT is therefore an adjacent evidence-selection principle, not a directly matched method or baseline for diagnosis-composition shift.

B. Multi-label shortcuts and unseen combinations

Multi-label models often exploit label dependence. Compositional multi-label classification explicitly holds out label combinations [3], while counterfactual and correlation-shortcut work studies harmful co-occurrence dependence [9], [10]. We

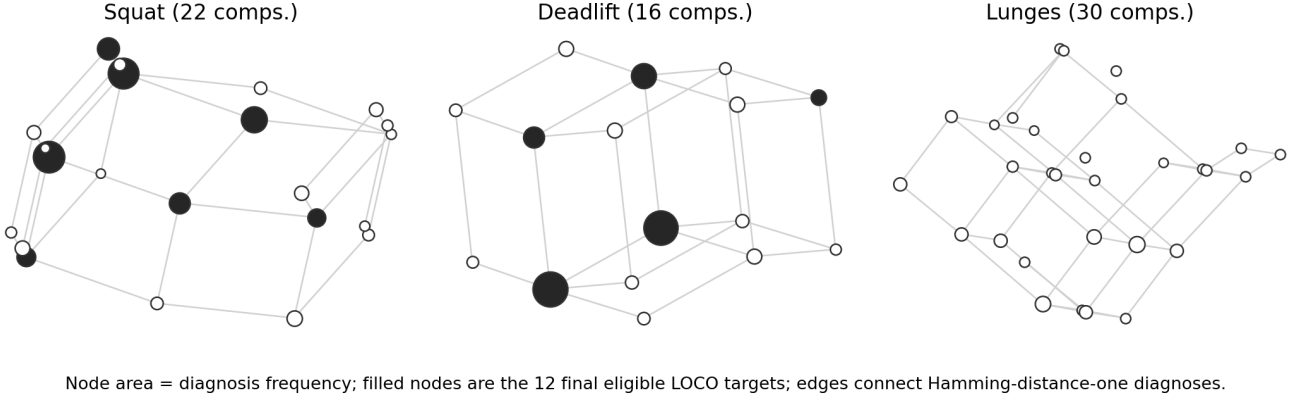


Fig. 1. Naturally observed diagnosis spaces. Node area represents repetition frequency, filled nodes are the 12 targets retained by the final support rule, and edges join diagnoses differing in exactly one criterion. Lunge has no filled node because its candidate targets cannot preserve the required train/validation state support after recording-group removal.

TABLE I
PROCESSED ALEX-GYM-1 DATA. “GROUPS” ARE PAIRED RECORDING IDENTIFIERS, NOT VERIFIED PARTICIPANT IDENTITIES.

Exercise	Repetitions	Criteria	Diagnoses	Groups
Squat	294	6	22	59
Deadlift	269	5	16	55
Lunge	106	7	30	53

use this literature to motivate a diagnostic, not a causal claim: label counts characterize exposure in the benchmark and never enter the exercise classifier.

III. DATA AND PROTOCOL

A. Dataset and preprocessing audit

The public ALEX-GYM-1 subset contains squat, Romanian deadlift, and lunge repetitions with frontal and lateral 33-joint 3D pose sequences. We convert each trainer criterion to an error bit: one denotes a violated criterion. Table I summarizes the processed data.

All-zero frames occur in both views: squat has 183/4,720 frontal (3.88%) and 702/4,720 lateral frames (14.87%); deadlift has 250/4,304 (5.81%) and 326/4,304 (7.57%); and lunge has 33/1,696 (1.95%) and 214/1,696 (12.62%), respectively. We linearly interpolate whole missing frames before pelvis centering, robust body-scale normalization, and resampling to 16 frames. One squat repetition has no valid frontal frame and is excluded. Raw-file SHA-256 hashes, the exclusion, and per-sequence quality counts are released. We do not impute isolated zero joints, whose semantics cannot be verified from the release.

B. Observed-composition holdout

For exercise e , let s_i identify a paired frontal/lateral recording and y_i its diagnosis. For target diagnosis y^* , the test set is all repetitions with $y_i = y^*$. Every recording group containing that target is removed from the train/validation pool, including its non-target repetitions. The remaining groups are split

deterministically. Candidate splits are rejected unless both states of every criterion occur at least eight times in training and at least once in validation. Train-validation, train-test, and validation-test group intersections are zero in every fold.

A target is eligible when it has at least ten repetitions, three source groups, and a feasible supported split. Seven squat and five deadlift diagnoses qualify. Two lunge diagnoses considered by the earlier protocol are infeasible: their minority pool contains exactly eight examples, so eight cannot remain in training while one also enters validation. They are excluded rather than evaluated under a different rule. The primary experiment is therefore 12 targets $\times$ three seeds per model.

The comparator is ordinary recording-group validation, not a guaranteed seen-composition set: 409 of 1,760 validation rows (23.2%) have a diagnosis absent from that fold’s training subset. RQ1 consequently measures ordinary grouped validation versus a *targeted* composition stress test; it does not attribute the entire gap solely to seen/unseen status.

C. Models and training

We use five compact controlled models: a temporal CNN, BiGRU, Transformer, selective state-space model, and Factorized Anatomical Criterion Tokens (FACT). All baselines receive a 16×198 sequence (two views $\times$ 33 joints $\times$ xyz). The TCN uses a 96-D frame embedding, temporal kernels 5 and 3, and mean/max pooling. BiGRU uses a 72-D bidirectional GRU; Transformer uses two 96-D, four-head encoder layers; and SSM uses one 32-D bidirectional selective scan. Their deadlift models contain 112k, 99k, 190k, and 13k parameters, respectively. The SSM is a compact study implementation rather than an official reproduction of a named large model.

FACT reshapes each sequence to $2 \times 16 \times 33 \times 3$. For criterion c , the annotation wording specifies one view and an initial joint set. A learned residual adjusts a sigmoid mask initialized near .95 on mapped joints and .05 elsewhere. Separate frontal/lateral encoders apply a 56-D frame projection, temporal convolutions of width 5 and 3, and mean/max pooling. A criterion-specific $112 \rightarrow 32$ adapter produces one token per criterion. Its logit

sums a local linear score and a scaled distance difference to two learned 32-D state prototypes. FACT never consumes other labels or logits at inference and contains 84,858 parameters for squat and 81,048 for deadlift. It is a locality-oriented comparator, not a claimed generally superior architecture.

Every model receives the same normalized two-view representation and split. Models use class-weighted binary cross-entropy, AdamW, fixed 0.5 thresholds, and early stopping on validation macro-F1. Hyperparameters are never selected using the held-out target. We report target-equal means: seeds are averaged within target, then all 12 targets receive equal weight. All primary trainers call the same deterministic constrained splitter. The released manifest records raw-file hashes, preprocessing version, exact indices, target support, exclusions, and zero group intersections for every target and seed.

IV. LABEL-SPACE DIAGNOSTIC

A. Label-Opposition Pressure

For criterion c , let $\mathbf{y}_{-c}^*$ denote the other criterion bits. Label-Opposition Pressure is

$$LOP_c(\mathbf{y}^*) = \sum_{i \in \mathcal{T}} \mathbb{I}[\mathbf{y}_{i,-c} = \mathbf{y}_{-c}^*] \mathbb{I}[y_{i,c} \neq y_c^*], \quad (1)$$

where $\mathcal{T}$ is the fold training set. Large pressure means many training examples reproduce the target context while supporting the wrong current bit. LOP is an exposure association, not causal influence or deployment confidence.

V. STATISTICAL ANALYSIS

The held-out diagnosis is the statistical unit. For RQ1, we average each metric over seeds and models within target, then test the 12 validation-LOCO differences. Exact match is primary; bit accuracy is secondary. For RQ2, we average TCN and FACT criterion accuracy over seeds, compute Spearman correlation within each diagnosis, and take the raw diagnosis-equal mean across the 12 informative diagnoses (seven contain six criteria and five contain five). A 50,000-shuffle test permutes pressure only among criteria of the same diagnosis. This blocked randomization retains within-diagnosis dependence and calibrates the chosen raw mean statistic without a parametric correlation assumption.

Both reported primary tests are two-sided. To avoid improving significance by removing a null result after inspection, their reported Holm values retain the original three-test family: the third audited analysis, a validation-fitted label-neighborhood deferral score, was inconclusive overall ($p = .092$) and is released as supplementary analysis rather than promoted as a contribution. We report 100,000-sample target bootstrap intervals.

VI. RESULTS

A. RQ1: targeted composition testing exposes exact-diagnosis failure

Figure 1 summarizes the target-level comparison. Averaged over models, ordinary grouped validation exceeds targeted

TABLE II
TARGET-EQUAL PERFORMANCE (%). VALIDATION IS ORDINARY GROUP-HELD-OUT; LOCO DELIBERATELY WITHHOLDS ONE NATURAL DIAGNOSIS.

Model	Validation			LOCO		
	Bit	μ F1	Exact	Bit	μ F1	Exact
TCN	65.08	56.35	12.61	63.04	44.86	2.39
BiGRU	65.32	57.00	14.71	60.61	42.11	2.63
Transformer	64.26	57.08	12.20	59.83	41.79	2.07
Selective SSM	66.89	56.91	14.23	62.50	41.82	3.41
FACT	72.39	63.95	18.56	64.89	44.77	5.14

LOCO exact match by 11.34 points (95% CI [8.09, 14.54]; two-sided Wilcoxon $p = .00049$, Holm $p = .00098$). Thus RQ1 is supported for its primary endpoint. Bit accuracy falls by 4.61 points (CI [0.25, 9.74]), but its two-sided $p = .151$ is not significant.

Table II shows that FACT has the largest mean bit accuracy and exact match, but its advantage over TCN is unresolved: +1.85 bit points ($p = .204$) and +2.75 exact points ($p = .086$) in two-sided exploratory comparisons. We therefore make no architecture-superiority claim. Table II reports micro-F1 pooled across criterion decisions. Macro-F1 is omitted because every criterion has one constant state inside a target fold; fold-level macro-F1 is not a meaningful discrimination measure in this protocol. “Bit” is the fraction of all criterion decisions that are correct; “Exact” requires every criterion of a repetition to be correct. Thus FACT’s 64.89 LOCO Bit but 5.14 Exact means that 64.89% of individual decisions are correct while only 5.14% of complete diagnosis vectors are correct.

B. RQ2: the one-bit label neighborhood localizes failure

Across 67 target-criterion units, LOP and held-out accuracy are negatively related within target. The raw diagnosis-equal mean Spearman correlation is $-.651$. The diagnosis-blocked two-sided permutation probability is $p < .00004$ (Holm $p = .00012$), supporting RQ2. Figure 2 shows the corresponding error trend. Because criteria within a diagnosis share samples and predictions, a pooled row-level parametric test would be anticonservative; the blocked permutation is the primary inference.

VII. DISCUSSION

The strongest result is not that a particular backbone wins; it is that exact multi-criterion diagnoses fail under a targeted natural composition stress test even when all criterion states retain training support. Marginal bit accuracy can hide this effect, explaining why its secondary gap is much smaller and statistically unresolved.

LOP supplies a simple account of which bit fails. If many training diagnoses match the held-out context but contain the opposite value of criterion c , a learner can associate the surrounding movement with the wrong bit. This remains an association: the classifier never consumes ground-truth co-labels at inference, and pose features may encode the same correlations.

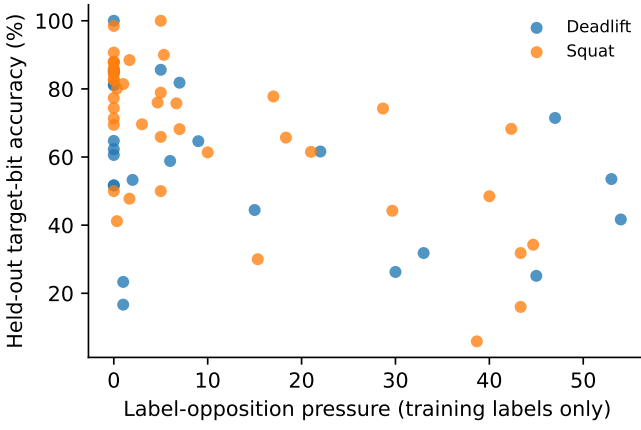


Fig. 2. One-bit training-label opposition versus LOCO criterion error for the TCN/FACT anchor analysis. Points are seed- and model-averaged target-criterion units; the line is descriptive.

VIII. LIMITATIONS

The study has one multi-label composition dataset and only 12 eligible diagnosis units. The split key identifies paired recordings, not verified participants, so we claim recording-pair rather than participant separation. Validation is not a pure seen-composition control. Manual criterion–anatomy mappings affect FACT, which is why FACT is only a comparator. Thresholds are fixed at 0.5, and no claim is made about calibrated probabilities. Missing-frame rates reach 14.87% in the squat lateral view; interpolation reduces but cannot eliminate uncertainty from pose-detector failures. Finally, exact match is clinically intuitive but harsh; it is reported alongside bit accuracy and micro-F1.

IX. CONCLUSION

Natural exercise diagnoses are more than collections of familiar bits. Under a recording-pair-disjoint observed-composition protocol, exact diagnosis match falls sharply relative to ordinary grouped validation. A one-bit training-label opposition statistic strongly identifies vulnerable criteria within diagnosis. The resulting contribution is a reproducible stress test and an honest label-space diagnostic, with clear evidence about both utility and limits.

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